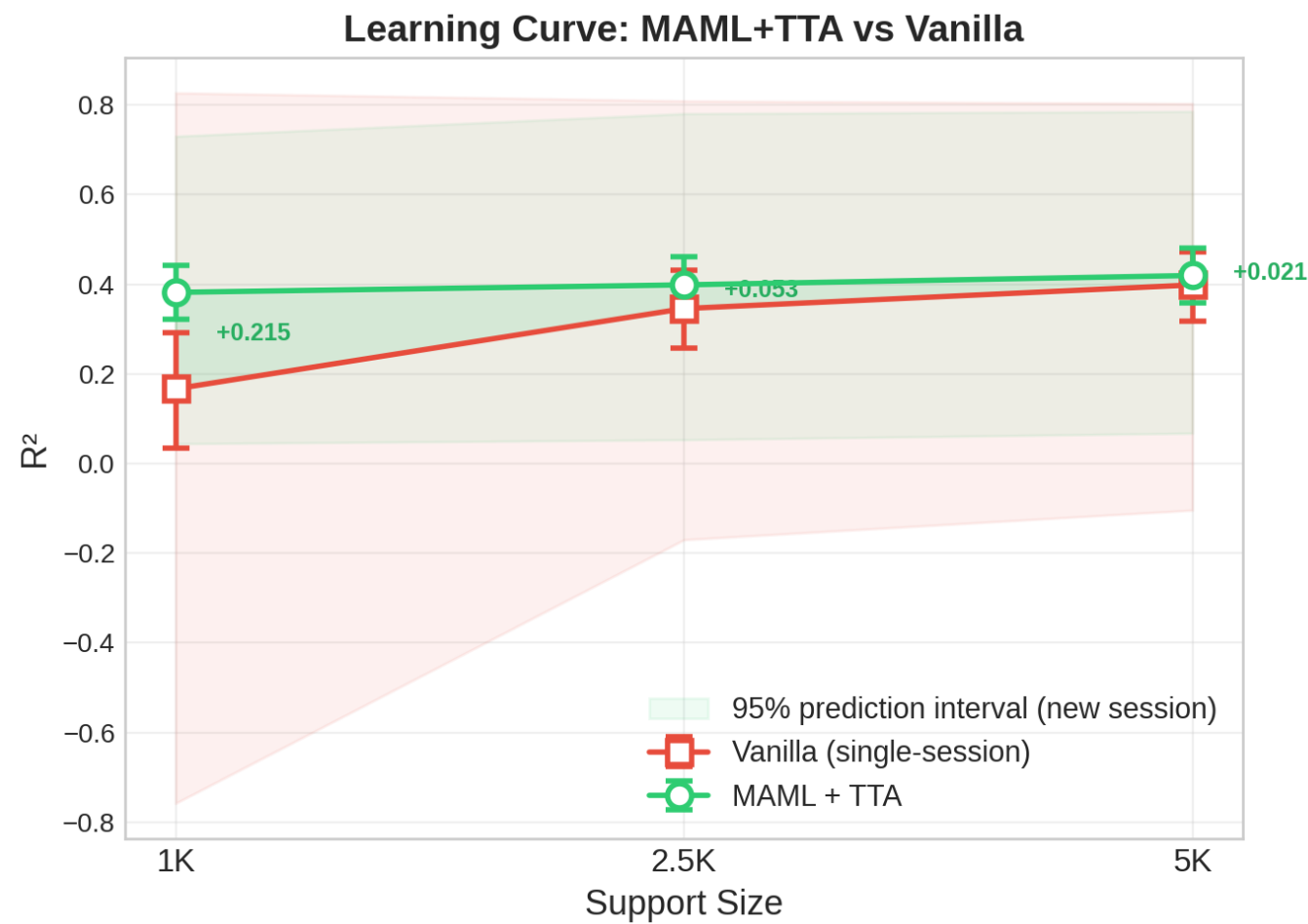


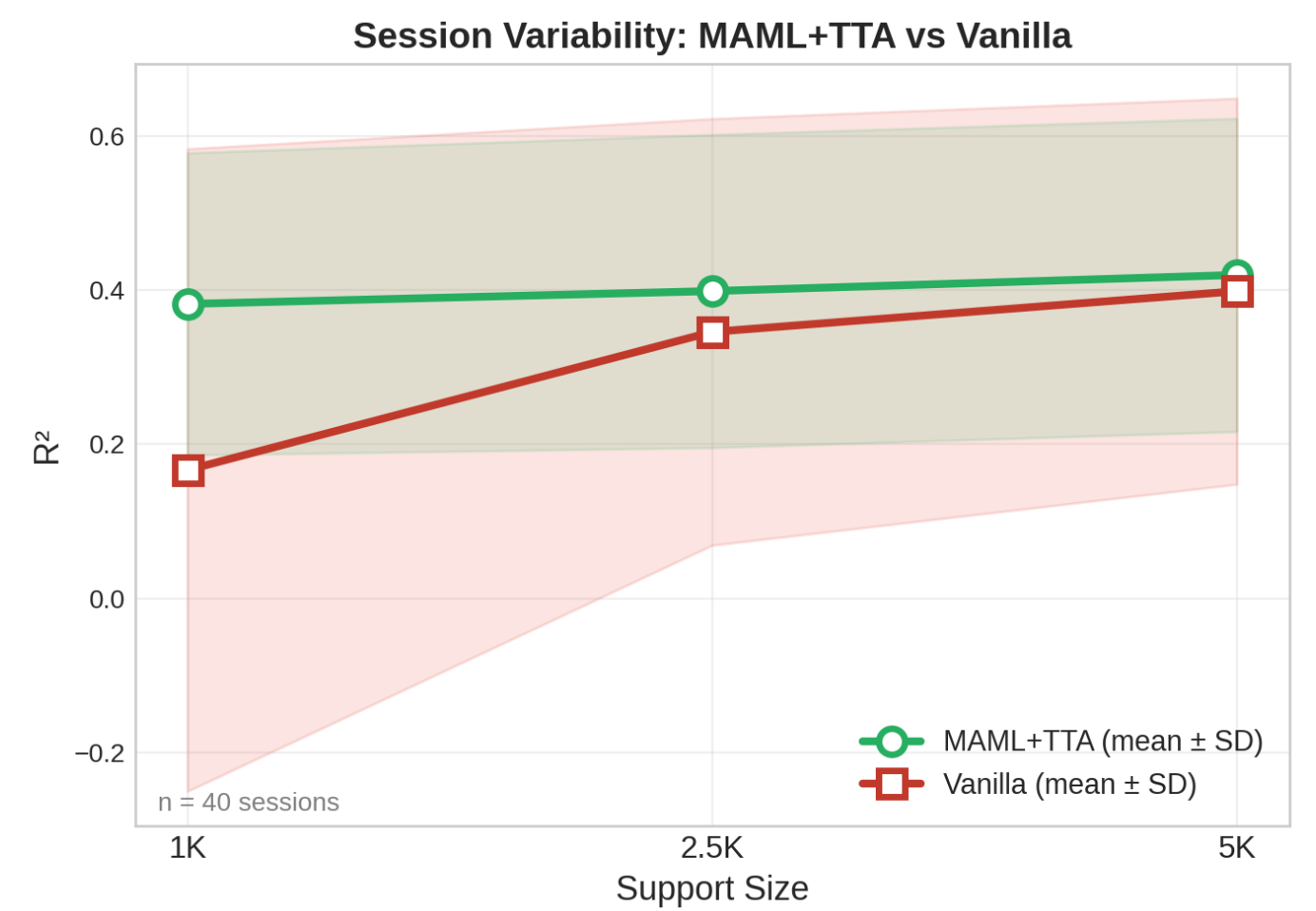
Bootstrap Methods in the Learning Curve Analysis

Figures

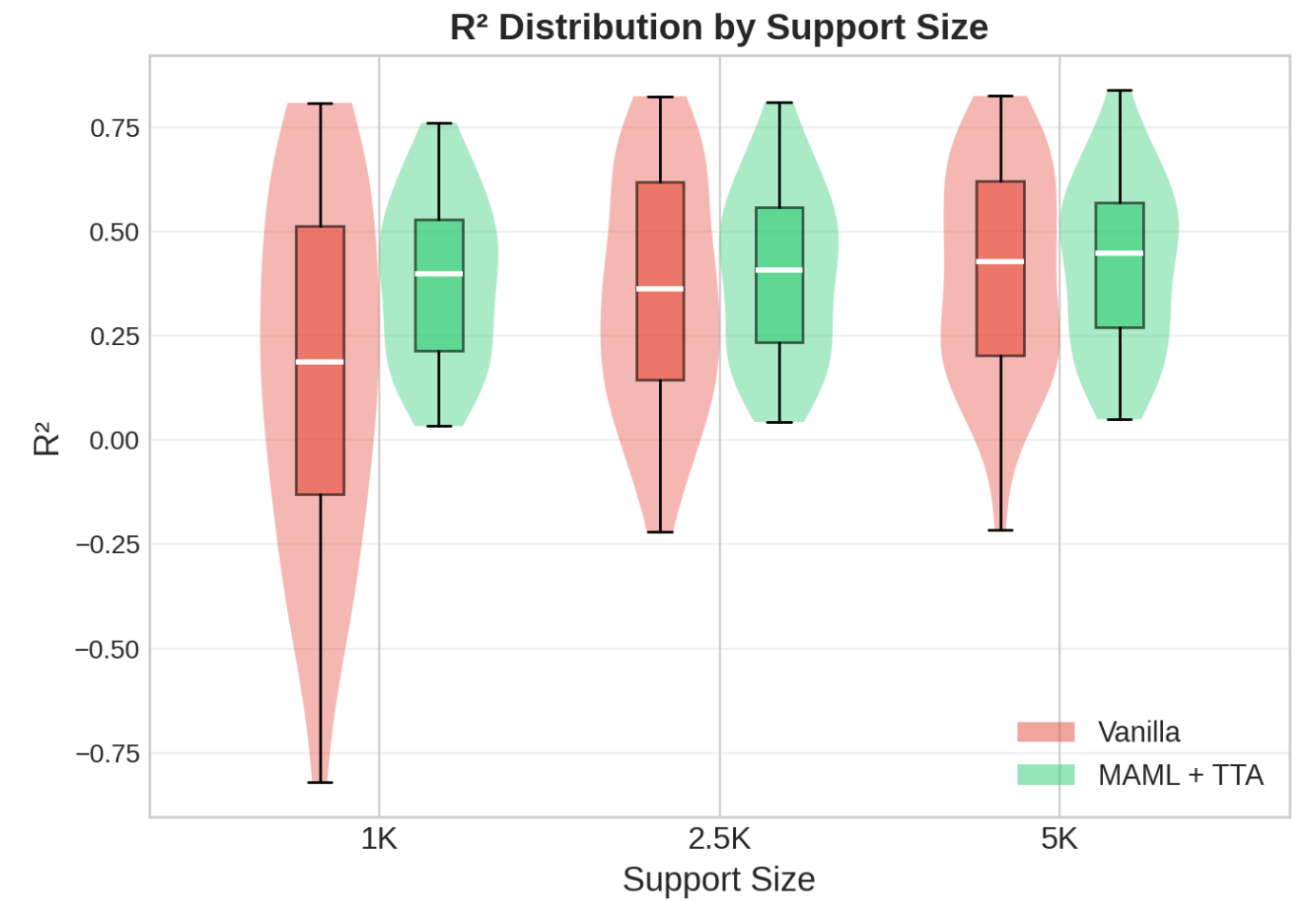
Learning Curve (mean $\pm$ 95% bootstrap CI, 95% prediction interval band)



Session Variability (mean $\pm$ SD)



R^2 Distribution by Support Size (violin + box)



Wilcoxon Signed-Rank Analysis

Wilcoxon Signed-Rank Analysis: 1K, 2.5K, 5K Support Samples



Computed Statistics (n = 40 sessions)

Bootstrap CI and Prediction Interval by Support Size

Support	Method	Mean	SD	95% CI	CI width	95% PI	PI width
1K	TTA	0.3819	0.1933	[0.3211, 0.4426]	0.1215	[0.0443, 0.7302]	0.6859
1K	Vanilla	0.1666	0.4111	[0.0334, 0.2917]	0.2584	[-0.7578, 0.8269]	1.5848
2.5K	TTA	0.3986	0.2003	[0.3367, 0.4612]	0.1245	[0.0530, 0.7807]	0.7277
2.5K	Vanilla	0.3457	0.2730	[0.2574, 0.4309]	0.1735	[-0.1699, 0.8090]	0.9789
5K	TTA	0.4195	0.2004	[0.3581, 0.4815]	0.1233	[0.0676, 0.7856]	0.7180
5K	Vanilla	0.3984	0.2470	[0.3177, 0.4725]	0.1548	[-0.1039, 0.8033]	0.9072

The CI is ~5–6× narrower than the PI at each support size, consistent with $SD / \sqrt{n} \approx SD / 6.3$.

The vanilla PI extends into negative R^2 at 1K and 2.5K support — reflecting sessions where vanilla training failed entirely. TTA's PI lower bound stays positive at all support sizes.

Wilcoxon Signed-Rank: TTA vs Vanilla

Support	TTA mean ± SD	Vanilla mean ± SD	Mean improvement	W	p-value	Effect r	Sessions improved
1K	0.3819 ± 0.1933	0.1666 ± 0.4111	+0.2153 ± 0.3492	144.0	0.000193 ***	0.6488 (large)	33/40 (82.5%)
2.5K	0.3986 ± 0.2003	0.3457 ± 0.2730	+0.0529 ± 0.2136	280.0	0.081714	0.3171 (medium)	25/40 (62.5%)
5K	0.4195 ± 0.2004	0.3984 ± 0.2470	+0.0211 ± 0.1632	347.0	0.404938	0.1537 (small)	23/40 (57.5%)

Pairwise TTA Comparisons (paired by session)

Comparison	Mean diff	W	p-value	Effect r	Sessions better
2.5K vs 1K	+0.0167	71.0	0.000001 ***	0.8268 (large)	35/40 (87.5%)
5K vs 1K	+0.0376	13.0	<0.000001 ***	0.9683 (large)	39/40 (97.5%)
5K vs 2.5K	+0.0209	3.0	<0.000001 ***	0.9927 (large)	39/40 (97.5%)

Diminishing Returns

Step	Mean gain ± SD
1K → 2.5K	+0.0167 ± 0.0177
2.5K → 5K	+0.0209 ± 0.0177

Wilcoxon comparing the two gains: $p = 0.735$ (ns). No significant diminishing returns — the gain from $2.5K \rightarrow 5K$ is similar in magnitude to $1K \rightarrow 2.5K$.

1. Bootstrap Confidence Interval on the Mean

What it estimates

The population mean R^2 — i.e., the expected R^2 if you recorded from infinitely many sessions drawn from the same population as your 40.

The math

Let $X = \{x_1, x_2, \dots, x_{40}\}$ be the observed session R^2 values at one support size. The sample mean is $\bar{x} = (1/n) \sum x_i$.

Bootstrap procedure (repeated $B = 10,000$ times):

1. Draw a resample $X^* = \{x_1^*, \dots, x_{40}^*\}$ from X with replacement
2. Compute the resample mean $\theta^*_b = \text{mean}(X^*)$
3. Store θ^*_b

CI construction (percentile method):

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CI = [ percentile( $\theta^*_b$ , 2.5%), percentile( $\theta^*_b$ , 97.5%) ]
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Why this works

The empirical distribution of X approximates the true population distribution F . Resampling from X mimics drawing new experiments from F . The spread of $\{\theta^*_b\}$ therefore approximates the sampling distribution of $\bar{x}$ under F — i.e., how much $\bar{x}$ would vary across repeated experiments of size 40.

Hypothesis / question answered

This is an **estimation** procedure, not a hypothesis test. The implicit question is:

"Where is the true population mean R^2 ?"

The CI answers: "With 95% confidence, the population mean R^2 lies within these bounds."

If the CI for TTA and the CI for vanilla do not overlap at a given support size, that provides informal evidence that the population means differ. However, the formal hypothesis test is the Wilcoxon signed-rank test (see below).

Assumptions

- The 40 sessions are representative of the population (sampling assumption)
- $n = 40$ is large enough for the empirical distribution to approximate F (asymptotic coverage)
- Does **not** assume normality or IID beyond exchangeability of sessions

What it does NOT account for

Sessions come from two monkeys (MonkeyG: 22, MonkeyJ: 18). Sessions within the same monkey are correlated. The bootstrap treats all 40 as exchangeable, which may slightly underestimate the true CI width. A cluster bootstrap (resampling at the monkey level) would handle this, but with only 2 monkeys there are too few clusters for it to be stable.

2. Bootstrap Prediction Interval

What it estimates

The range of R^2 values where a **single new session** drawn from the same population would likely fall — not where the average would land, but where one individual observation would.

The math

Bootstrap procedure (repeated $B = 10,000$ times):

1. Draw a resample $X^* = \{x_1, \dots, x_{40}\}$ from X with replacement
2. Compute the resample mean $\mu^* = \text{mean}(X^*)$
3. Draw one residual $\epsilon \sim \text{Uniform}(X - \bar{x})$ [a randomly chosen deviation from the original mean]
4. Compute the simulated new observation: $\tilde{y} = \mu^* + \epsilon$
5. Store $\tilde{y}$

PI construction:

$$\text{PI} = [\text{percentile}(\tilde{y}, 2.5\%), \text{percentile}(\tilde{y}, 97.5\%)]$$

Why $\tilde{y} = \mu^* + \epsilon$

This construction captures two sources of variability:

- μ^* : the uncertainty in where the population mean is (same as the CI above)
- $\epsilon = x_i - \bar{x}$: the natural spread of individual sessions around the mean

A new session's $R^2 \approx$ population mean + individual deviation. The bootstrap approximates both. Adding them together gives a simulated draw of a new session's R^2 .

Comparison with CI

The prediction interval is always wider than the CI. For large n :

CI width $\propto$ SD / $\sqrt{n}$	(shrinks as you get more sessions)
PI width $\approx 2 \times$ SD	(stays wide regardless of n – irreducible variability)

With $n = 40$ and TTA SD ≈ 0.20 , the observed CI widths are ~ 0.12 and PI widths are $\sim 0.69\text{--}0.73$ — approximately a $6\times$ ratio, matching the theoretical $SD / (SD/\sqrt{n}) = \sqrt{40} \approx 6.3$ prediction.

3. Wilcoxon Signed-Rank Test (the actual hypothesis test)

The bootstrap intervals are descriptive. The Wilcoxon tests are the formal inferential tests.

Setup

For each support size, compute per-session differences:

$$d_i = \text{TTA } R^2_i - \text{Vanilla } R^2_i \quad \text{for } i = 1, \dots, 40$$

Null hypothesis

$$H_0: \text{The distribution of } d_i \text{ is symmetric around zero.}$$

Equivalently: TTA and vanilla are equally good; any observed difference is due to chance.

Alternative hypothesis

$$H_1: \text{The distribution of } d_i \text{ is shifted away from zero (two-sided).}$$

The test statistic W is the sum of ranks of positive differences. Under H_0 , W follows a known distribution. A small p -value means the observed pattern of differences is unlikely under symmetry around zero.

Why Wilcoxon and not a t-test

The t-test assumes the differences d_i are normally distributed. With $n = 40$ sessions spanning two monkeys and two brain areas, the distribution of d_i may be heavy-tailed or bimodal. The Wilcoxon test only requires that d_i values are exchangeable under H_0 (i.e., that positive and negative differences are equally likely), which is a much weaker assumption.

Summary Table

Technique	Question answered	Width	Assumes IID?
Bootstrap CI on mean	Where is the population mean R^2 ?	$\sim 0.12\text{--}0.17$	No
Bootstrap prediction interval	Where will the next individual session land?	$\sim 0.69\text{--}1.58$	No

Technique	Question answered	Width	Assumes IID?
Wilcoxon signed-rank test	Is TTA significantly better than vanilla?	p-value	No

Reading the Learning Curve

- **Line:** sample mean R^2 across 40 sessions
- **Error bars** (narrow, capped): 95% bootstrap CI — precision of the mean estimate
- **Shaded band** (wide, faint): 95% prediction interval — expected range for a new session
- **Annotated gap:** mean R^2 improvement (TTA – vanilla) at each support size

At 1K support TTA is significantly better ($p < 0.001$, large effect). At 2.5K and 5K the mean gap persists but is not significant — the prediction interval and CI overlap between TTA and vanilla is consistent with Wilcoxon $p = 0.08$ and 0.40 respectively. More support data closes the gap because vanilla performance improves faster than TTA performance.